Information Constraints in Fine-tuning and Reasoning Language Models

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Abstract

This thesis reports the work I carried out during my 2026 internship with the MILeS group at LAMSADE, as part of the completion of my IASD M2 degree, under the supervision of Paul Caillon and Alexandre Allauzen. The main topic of the internship was the study of information-theoretic limits on quantized fine-tuning and reasoning in large language models (LLMs). This report is organized in two main parts, each adressing a specific research question. The first part focuses on the theoretic limits of quantized-finetuning via the study of information contained in LoRA/QLoRA adapters. The second part focuses on the study of reasoning trajectories and the information contained in reasoning states, in the context of reasoning language models (RLMs).

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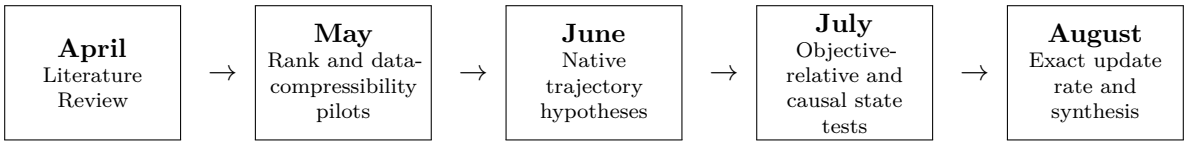


Figure 1: Rough chronology of the internship so far, which began on April 13th and is set to end on September 28th.

1 Introduction

1.1 Internship context

Having seen the initial offer for the internship posted by my tutors, Paul and Alexandre, I was enthusiastic to start working on it, emailed Alexandre, had an interview with Paul and another with him before they decided to move forward. The internship is taking place from April to September 2026 under the Université Paris Dauphine–PSL. I'm working in practice from the PariSantéCampus offices, and interacting daily with other M2 students, PhD students, and researchers. Under my role of M2 intern, I am part of *MILES (Machine Intelligence and Learning Systems)*, a research group within *LAMSADE*, the *Laboratoire d'Analyse et de Modélisation de Systèmes d'Aide à la Décision*, a joint research laboratory of Université Paris Dauphine–PSL and the CNRS.

LAMSADE is Dauphine's main computer science and decision-science research laboratory, spanning decision support, optimization, algorithms, data science, and artificial intelligence, while MILES focuses more specifically on the theoretical and algorithmic foundat

1.2 Research trajectory

The internship began with an information-theoretic study of fine-tuning, where early capacity and dataset-compressibility experiments exposed the need for an operational definition of update rate. While doing the literature review, I had some interesting parallel ideas which triggered an investigation of reasoning trajectories, testing whether language models expose compact, reusable intermediate states, which increasingly showed that useful state descriptions depend on the downstream objective. These results motivated more causal state-interface experiments and, in parallel, a return to fine-tuning with exact serialized-rate measurements. The two lines eventually converged on a common question: how much information must a representation preserve, relative to a particular receiver and utility, for learned behavior to survive? At the time of writing, this question is still ongoing with some promising results, and we're looking to apply it to reasoning to use results from both of the tracks we've pursued so far.

1.3 Structure of the report

This report's chapters follows a chronological order with respect to which research directions were tackled first. Given that the goal of this report is as much to highlight our researching process as to explain the results this process yielded, negative results remain in it, as they often determined the next experiment to be ran, if not a high-level course adjustment. We begin by with a brief exploration of the technical background to the research we led, including an overview of the relevant concepts. Then, before each of the two main sections (respectively information-theoretic limits of finetuning and reasoning trajectories), we give an overview of the literature, before explaining how they contributed to motivating our initial research questions.

1.4 Research questions

1. How should the rate of a fine-tuning update be defined when the receiver already has the base model, adapter layout, and decoder?
2. How much serialized update information preserves learned behavior, and which data or model properties predict that threshold, especially in the context of LoRA finetuning?
3. In the context of CoT-generating RLMS, do reasoning trajectories admit a canonical latent-space based decomposition? If not, how variable are the decompositions across different utilities?
4. When can an explicit state code support RLM causal handoff and reliable recursive use?

1.5 Contributions

Our main contributions include:

1. **Weight-space fidelity does not determine preservation of learned behavior.** LoRA adapters with lower reconstruction error, e.g. via RMSE, can retain substantially less of the learned behavior than adapters with higher reconstruction error, showing that ordinary weight distortion is not an adequate measure of useful information. (Section 3.4)
2. **The accuracy gain of a model over a given dataset is a very poor predictor for the necessary amount of information of its fine-tuning update.** Across tasks, large behavioral improvements can correspond to very small behavior-preserving adapters: most notably, XBRL produces the largest fine-tuning gain in the campaign while requiring the lowest measured rate. (Section 3.7)
3. **The information written by fine-tuning is best expressed as relative to the frozen base model.** The same datasets require different adapter rates for different base models, and initial results suggest that model-relative properties of the corrections requested by the data predict this rate better than dataset size, base-model loss, or prediction gain alone. (Section 3.7)
4. **Chain-of-thought segmentation is strongly utility-dependent.** Boundaries that are useful for answer formation, symbolic updates, correctness, or compression differ substantially; no single segmentation performs well across all objectives. (Section 4.3)
5. **In RLMs, decodable intermediate states are not necessarily reusable causal states.** Native reasoning states could often be identified or decoded, but attempts to reuse them reliably in downstream computation were much weaker, suggesting that reasoning is not naturally organized as a sequence of clean, independently reusable states. (Section 4.6)
6. **Explicit state interfaces can at least weakly support reliable long-horizon reasoning.** When a compact state representation is given a stable meaning and trained to remain closed under repeated updates, it can support substantially more reliable recursive computation than one-pass reasoning. (Section 4.7)

2 Technical background

This section introduces the general concepts and notation needed for the rest of the report, and can be skipped by readers already familiar with the state of the art of reasoning language models, parameter-efficient fine-tuning, and representation analysis. The more specialized recent literature that directly motivates our experiments is deferred to the literature reviews in Sections 3 and 4.

2.1 Transformers and autoregressive language models

A language model assigns probabilities to sequences of discrete tokens. Modern large language models (LLMs) typically tokenize text into subword units, map each token to a vector representation, and process the resulting sequence through many transformer blocks (Vaswani et al. 2017). Each block combines self-attention, which lets a position aggregate information from earlier positions, with feed-forward transformations. Residual connections carry a vector-valued representation through the successive layers of the network.

For a prompt x and generated tokens $y_{1:T}$, an autoregressive model factorizes the probability of the continuation as

$$p_\theta(y_{1:T} \mid x) = \prod_{t=1}^{T} p_\theta(y_t \mid x, y_{<t}). \tag{1}$$

At each position, the final hidden representation is mapped to logits over the vocabulary and then to a probability distribution for the next token. Generation repeatedly samples or selects from this distribution and appends the chosen token to the context.

We write $h_t^{(\ell)} \in \mathbb{R}^d$ for the hidden representation at token position t and layer ℓ. For a fixed generated sequence, the collection $(h_1^{(\ell)}, \ldots, h_T^{(\ell)})$ can be viewed as a trajectory through representation space. This latent trajectory is distinct from the visible text sequence: two similar sentences can be represented differently internally, and a large internal change need not coincide with a linguistic boundary.

2.2 Chain-of-thought and reasoning traces

Chain-of-thought (CoT) prompting asks a language model to produce intermediate reasoning steps before its final answer. This simple change can substantially improve performance on multi-step reasoning tasks (Wei et al. 2022) and problem-solving more broadly. Models trained or prompted to produce such traces are often called reasoning language models (RLMs) in this report. Since 2022, however, chain-of-thought has evolved from primarily a prompting technique into a behavior deliberately induced during training. Methods such as reinforcement learning from verifiable rewards (RLVR) and optimization algorithms such as Group Relative Policy Optimization (GRPO) train models directly against the correctness of their final answers and can induce increasingly long, self-correcting reasoning traces without requiring explicit step-by-step demonstrations at inference time (DeepSeek-AI 2025; Shao et al. 2024). Modern reasoning models therefore commonly generate extended chains of thought by default, with the amount and structure of intermediate computation becoming part of the learned policy rather than merely a consequence of prompt engineering. Post-training can also give CoT a distributional form quite unlike ordinary language generation: the trace is optimized primarily as an internal computational workspace rather than as communicative prose, which can make stylistic or formatting interventions brittle. Although it is produced by the same autoregressive mechanism, this different functional role gives reason to expect correspondingly different latent dynamics.

A CoT trace gives two related objects to study. The first is the written trajectory: the sequence of generated tokens, sentences, equations, or explicit intermediate claims. The second is the latent trajectory formed by the model's hidden representations while generating those tokens. The written trace is directly observable, while the latent trajectory requires access to model activations. Nothing in the autoregressive architecture requires meaningful computational steps to coincide with sentences, punctuation, or any other fixed textual unit.

2.3 Fine-tuning, parameter efficiency, and LoRA

Pretraining produces a general-purpose parameter vector θ_0. Fine-tuning adapts that model to a new dataset or task by continuing optimization on a new objective. Where full fine-tuning updates most or all model parameters, parameter-efficient fine-tuning instead keeps most of the pretrained model fixed and learns a much smaller task-specific object.

Low-Rank Adaptation (LoRA) is a widely used example (Hu et al. 2022). For a pretrained weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA freezes W and learns a low-rank update

$$W' = W + \Delta W, \qquad \Delta W = \frac{\alpha}{r} BA, \tag{2}$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$, and r is much smaller than the dimensions of W. LoRA modules can be inserted into several projections of each transformer block while the underlying pretrained weights remain unchanged.

This gives a useful separation between a *base model*, which is shared, and a comparatively small learned *adapter*. Rank controls the dimension of the update parameterization, but it should not be confused with a number of information bits: the learned matrix entries still need numerical values, and many different numerical representations are possible at the same rank.

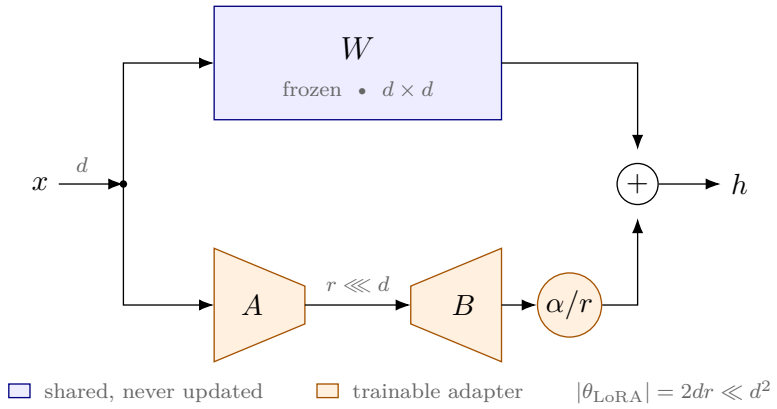


Figure 2: LoRA freezes the pretrained weight matrix W and learns the low-rank adapter update $\Delta W = (\alpha/r)BA$.

2.4 Quantization and QLoRA

Neural-network parameters are usually stored as floating-point numbers. Quantization represents them with a smaller finite set of values. A simple quantizer stores, for example, an integer code for each weight together with one or more scale parameters that map the codes back to approximate real values. Fewer available levels reduce memory use but increase numerical distortion.

Quantization can be applied to different objects for different reasons. A pretrained backbone may be quantized to reduce memory during inference or fine-tuning, while a separately learned adapter may remain in higher precision. Representative methods include GPTQ (Frantar et al. 2023), ZeroQuant (Yao et al. 2022), SmoothQuant (Xiao et al. 2023), and AWQ (Lin et al. 2024) for post-training or deployment quantization. QLoRA uses the same separation during adapter training: it keeps a pretrained model in a four-bit quantized representation while training LoRA adapters through it (Dettmers et al. 2023). Its NormalFloat-4 (NF4) representation is designed for weight distributions encountered in pretrained networks.

2.5 Information, coding, and rate–distortion

Information theory provides a language for relating probability, coding, and compression. If an event x has probability $p(x)$, its surprisal in bits is

$$-\log_2 p(x). \tag{3}$$

The entropy of a random variable is the expected surprisal. In an ideal code, more probable events can be represented with shorter descriptions, so negative log-probability can also be interpreted as an idealized code length. For a language model, average negative log-likelihood therefore measures both predictive error and the number of bits per token that the model would need in an ideal probabilistic code.

Conditional quantities express the value of information when some side information is already known. If both sender and receiver already possess a pretrained model, for example, our interest would lie in the number of additional bits needed to specify whatever changes relative to it, rather than how many bits are required to describe that model again. *Mutual information* similarly measures how much knowing one random variable reduces uncertainty about another.

Rate–distortion theory asks how small a representation can be while keeping some notion of error below an acceptable level (Cover and Thomas 2006). The key point is that a rate has meaning only together with a distortion measure. Two compressed objects with the same numerical reconstruction error can behave differently if errors occur in different functionally important directions. Conversely, a representation may differ substantially in Euclidean space while preserving the behavior that matters.

Minimum Description Length (MDL) uses description length as a way of expressing regularity: a model or explanation is useful when it gives a short joint description of the model and the observations (Grünwald and Roos 2019). In this report, MDL and rate–distortion ideas provide conceptual motivation, while the exact rate and distortion that ended up being most widely used for the LoRA experiments are defined in Section 3.

2.6 Internal representations, probes, and causal interventions

As previously introduced, by "hidden representations", we describe the vectorial state of a transfomer pass after any one of its Attention+MLP layers. Hidden representations can contain information that is not explicit in the generated text. A common way to test this is a *probe*: a simple supervised model, often linear, is trained to predict a property of interest from hidden activations. High held-out probe performance shows that the property is statistically accessible from the representation under the chosen decoder.

Geometric analyses ask related but unsupervised questions. Principal Component Analysis (PCA) can identify high-variance directions; cosine similarity measures changes in direction; distances or local statistics can reveal how a trajectory evolves through representation space. Such measurements describe organization in the representation but do not by themselves show that the model uses that organization causally.

Where probes offer evidence for decodability, causal interventions provide stronger evidence. In activation patching or interchange experiments, an internal activation from one computation is replaced by an activation from another and the downstream output is observed. If changing a representation changes later behavior in the way predicted by its proposed semantics, this supports a causal role. It is therefore useful to keep three claims separate: a variable may be *decodable*, may form a clear *geometric structure*, and may be *causally used*; importantly, none of these necessarily implies the others.

2.7 Experimental conventions

For language modeling, we frequently report held-out negative log-likelihood (NLL), often converted to bits per token using base-two logarithms. This is a dense metric as every scored token contributes an observation. We also resort to exact or regex-based string matching for tasks with a small number of possible outputs, such as multiple-choice questions or arithmetic problems. In these cases, we report the fraction of correct answers.

Controlled comparisons also require matching the resource that is meant to stay fixed. Depending on the experiment, this can mean the same number of optimizer updates, examples, tokens, parameters, or representational capacity. We openly state the matching rule with each experiment rather than assumed from a shared nominal setting.

3 Fine-tuning under an information budget

3.1 Literature review

3.1.1 Low-dimensional adaptation

Intrinsic-dimension results show that successful language-model adaptation can lie in a much smaller subspace than the full parameter space (Aghajanyan et al. 2021). The original LoRA method leverages that constraint, by learning a low-rank update to selected weight matrices (Hu et al. 2022). On top of LoRA, AdaLoRA allocates rank by an importance score rather than fixing the same rank throughout the model (Zhang et al. 2023). Work on LoRA factor asymmetry further shows that the input and output factors can contribute unequally despite having the same rank and similar storage cost (Zhu et al. 2024).

3.1.2 Quantization and quantization-aware adaptation

Whole-model quantization provides several ways to allocate error. GPTQ uses approximate second-order information during sequential weight quantization (Frantar et al. 2023). ZeroQuant uses hardware-aware groups and layer-wise distillation (Yao et al. 2022); SmoothQuant moves activation difficulty into weights before applying integer quantization (Xiao et al. 2023); and AWQ preserves channels selected from activation statistics (Lin et al. 2024). These methods show that equal nominal width does not imply equal functional cost across weights or layers.

QLoRA trains higher-precision adapters over a four-bit frozen backbone (Dettmers et al. 2023). LoftQ chooses an initialization that reduces joint quantization and low-rank approximation error (Li et al. 2024), while QA-LoRA aligns quantization groups with the adapter so that the learned change can merge into a low-bit model (Xu et al. 2024). Their main targets are training memory, initialization error, or low-bit deployment. They do not directly measure the shortest complete file that reproduces the behavior of one fixed trained adapter.

3.1.3 Description length, bounds, and open measurement

Rate–distortion theory formalizes the least message rate compatible with a chosen distortion (Berger 1971; Cover and Thomas 2006). Minimum Description Length (MDL) treats short joint descriptions as a model-selection principle (Grünwald and Roos 2019). Several more recent results make these ideas concrete in neural networks.

Young (2025) takes the rate–distortion view literally in *Radio*, framing post-training LLM quantization as a trade-off between the number of transmitted bits and the distortion induced by compression. Their method can target either a desired compressed model size or a tolerated

loss of accuracy. Their framing of compression as an allocation problem under a global rate budget highlights that a bit need not have the same functional value everywhere in a network.

Liu et al. (2025) give a complementary empirical picture with *ParetoQ*, a unified framework for comparing 1-, 1.58-, 2-, 3-, and 4-bit quantization. They report a marked transition between two and three bits: at three bits and above, fine-tuned quantized models remain comparatively close to the pretrained distribution, whereas at two bits and below the learned representations change much more substantially. Their size–accuracy results also show that lower nominal precision does not translate monotonically into worse trade-offs. This hints at bits per parameter not being a linear measure of usable model capacity; and at the behavioral meaning of a rate depending on representation and training regime.

A different connection between compression and learning appears in the older work of Arora et al. (2018). They construct explicit succinct reparameterizations of trained deep networks and use them to obtain generalization bounds substantially stronger than naive parameter counting. They rely on noise-stability properties, that let the compressed network approximately preserve the original predictions. Later PAC-Bayes work develops related compression-based bounds (Zhou et al. 2019; Lotfi et al. 2022).

Finally, Bergsträßer et al. (2026) study description length from the opposite direction in *Transformers are Inherently Succinct*. In a formal-language setting, they prove that fixed-precision transformers can represent some languages exponentially more succinctly than linear temporal logic or recurrent networks, and doubly exponentially more succinctly than finite automata. Though not about fine-tuning, this theorem gives a useful intuition for our setting: the frozen transformer is a powerful decoder rather than passive side information, and a small update may select or specify a behavior whose standalone description would be far larger. This makes the relevant quantity a description *relative to the base model* rather than an intrinsic description of the task.

3.2 Initial pilots

Throughout this section, a training set is a collection of supervised prompt–response *examples*. We call one model–dataset–training condition an experimental *arm*, and a related collection of arms a *panel*.

Once our main literature review phase was over, we wanted to use a cheap measurable proxy for how much a fine-tune writes. The LoRA's matrices' rank was the cheapest candidate, as if useful adaptation really occupies a small subspace, the smallest effective rank might behave like a crude information budget. We therefore began with a small QLoRA rank sweep before building a true codec.

An initial Qwen2.5-0.5B sweep used instruction-following examples from the *Alpaca* and *Dolly* datasets, with LoRA ranks $\in \{0, 2, 4, 8, 16\}$; rank 0 being the frozen-model baseline. Most improvement in held-out next-token loss appeared between rank 0 and rank 2, with smaller gains thereafter. Useful adaptation occupied a small low-rank subspace in these runs.

On top of rank, we also wondered whether the amount or ordinary compressibility of the training data predicted how much information would be written. Our pilot dataset variants either duplicated examples or made their prompt–response templates more regular. Exact duplication is deliberately content-free, as it increases the nominal number of stored examples without adding a new training case. But under epoch-based training, it also changes optimization exposure and revisiting frequency.

Separately, A-only (with A being the down-projecting matrix of the LoRA adapter) LoRA training stayed near baseline, while B-only recovered much of full-LoRA performance in the tested setup. This functional asymmetry warned against treating equally sized parts of the adapter as equally valuable.

Though they carried a lot of negative results and imprecisions, these initial pilots helped

us refine our intuition and eliminate a lot of easy targets: nominal rank is not a rate, nominal example count is confounded with training exposure, superficial corpus compressibility is not a clean information intervention, and a small exact-accuracy benchmark is too noisy. Our next methodological step was therefore to hold one learned update fixed and vary only how many bits were used to store it.

3.3 Fixed-checkpoint exact-rate protocol

To measure an actual rate–distortion curve, we train one rank-16 LoRA adapter and then keep that checkpoint fixed. Every point on the curve is therefore a different compressed representation of the same learned update, rather than a separately trained model.

Our codec quantizes the LoRA factors, bit-packs the quantized values together with their scales and metadata, compresses the payload with zlib, and writes the result to disk. We then reload that file into the frozen base model before evaluation. The reported rate is the complete serialized file size divided by the number of values in the original adapter, in bits per adapter value.

At one bit and above, each LoRA matrix is quantized row by row. At one bit, a value is represented only by its sign and its row scale, so it reconstructs to $+\operatorname{mean}|w|$ or $-\operatorname{mean}|w|$. Higher rates use more quantization levels. Below one bit per value, we cannot give every value even one bit, so we instead keep only a fraction of LoRA rank directions, always keeping or dropping the corresponding directions in A and B together.

We measure behavior mainly with held-out next-token negative log-likelihood. If L_0 is the frozen-model loss, L_{raw} the loss of the uncompressed adapter, and L_r the loss after compression to rate r, retained gain is

$$\operatorname{Retain}(r) = \frac{L_0 - L_r}{L_0 - L_{\text{raw}}}.$$

We use $R^\star(.90)$ for the smallest measured rate that retains at least 90% of the uncompressed adapter's gain. The 90% threshold is simply a consistent operating point for comparing experiments; the full rate–retention curves remain available when the threshold hides useful detail.

3.4 Adaptive allocation: a negative result

Once the rate axis was defined, we asked whether a fixed bit budget could be allocated more intelligently than uniform precision. Weight reconstruction error was a deliberately simple first proxy: if the rows that are hardest to reconstruct are also the rows whose corruption matters most for behavior, an error-aware allocator should win at the same file size.

The adaptive codec could drop each row of a LoRA factor matrix or encode it at 1, 2, 3, 4, or 8 bits. A Lagrangian objective traded reconstruction error against code length and chose the allocation. Contrary to the hypothesis, this did not beat uniform coding. In the clearest counterexample, an adaptive point had lower adapter RMSE than a fixed-rate point but preserved far less of the learned behavior.

This result ruled out global weight error as the main quantity we wanted. It also shifted the question from *where* to spend a fixed budget to what determines how large the required budget is in the first place.

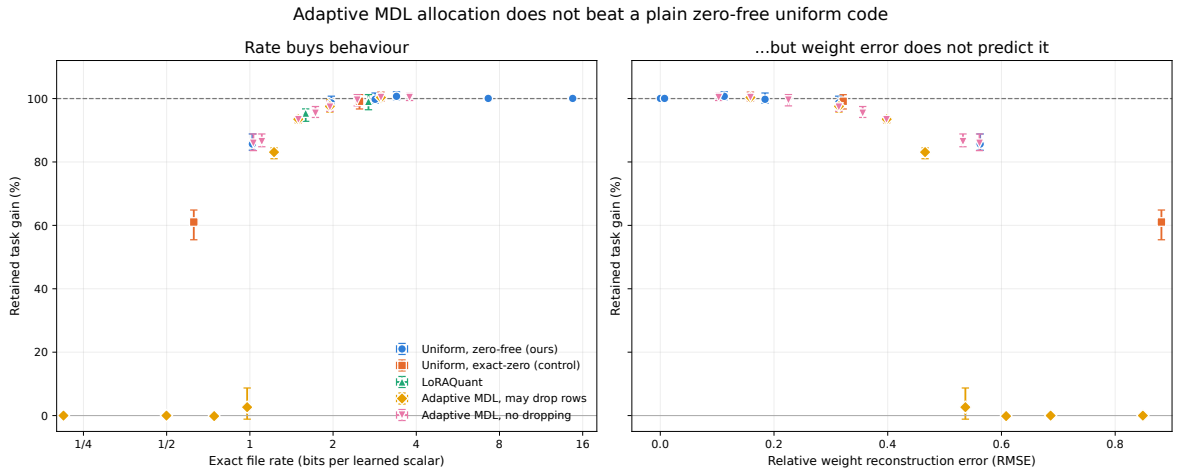


Figure 3: Adaptive MDL allocation against fixed-rate controls. Left: retained held-out prediction gain against exact decoded file rate. Right: retained gain against relative weight RMSE.

3.5 Distinct examples at fixed compute

We first tested whether the required rate grows with the amount of distinct training data. A naive dataset-size sweep would also change training exposure, so we varied the number of distinct MetaMathQA examples from 2,000 to 32,000 while holding both optimizer steps (2,000) and total example presentations (32,000) fixed.

Distinct examples	Epochs	$R^\star(.90)$	Seed range	Raw GSM8K EM
2,000	16	0.645	0.611–0.700	0.546
4,000	8	0.740	0.712–0.787	0.546
8,000	4	0.709	0.671–0.747	0.618
16,000	2	0.832	0.773–0.908	0.676
32,000	1	1.002	0.983–1.015	0.694

Table 1: $R^\star(.90)$ in serialized bits per raw adapter value, with optimizer steps and training-example presentations fixed.

Across the full range, $R^\star$ predictably generally increases with the number of distinct examples: the fit $R^\star = -0.26 + 0.081 \log_2 n_{\text{distinct}}$ gives $R^2 = 0.78$ over 15 runs. Though the trend is not perfectly monotone, it can be used as a rough scaling law. In the most repeated conditions, compression can even outperform the raw adapter on held-out loss, suggesting that it removes some overfit components. This is therefore a result that supports a broad relationship between distinct training data and required adapter rate.

Distinct examples, however, are only a rough proxy for distinct content. Two examples can be different records while still expressing the same underlying problem. We therefore ran a second experiment with exactly 8,000 training examples in every condition, but varied how many source math problems those examples came from. Across six seeds, increasing the number of source problems from roughly 400 to 5,620 raised mean $R^\star$ from 0.682 to 0.742 bits per value.

This made content diversity look more promising than example count alone, but another confound remained: the adapters trained on more diverse data also tended to improve the model more. Retaining 90% of a larger improvement may itself require a larger file. We therefore turned to experiments that could separate source information from the amount of behavior learned.

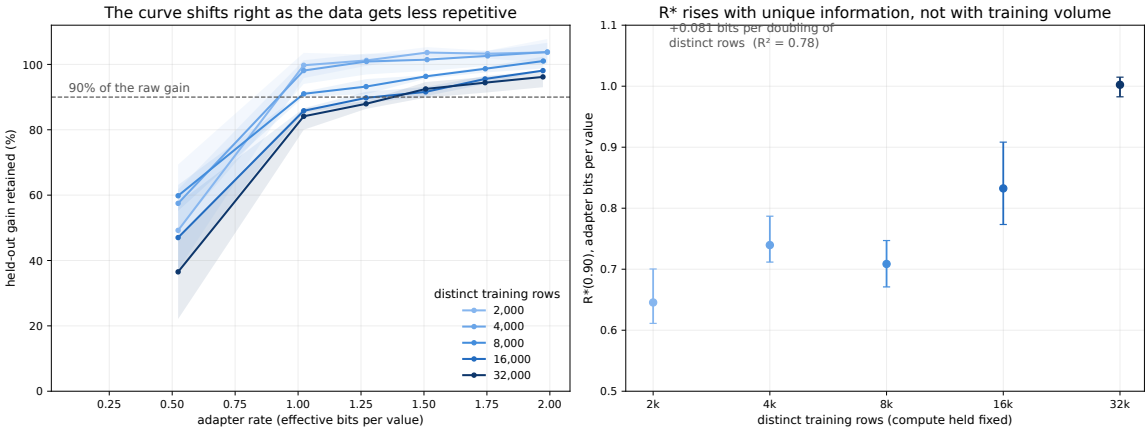


Figure 4: Compute-matched distinct-example experiment. The left panel shows the full rate–retention curves; the right panel shows the derived $R^{\star}(.90)$ cells and the fit over 15 runs.

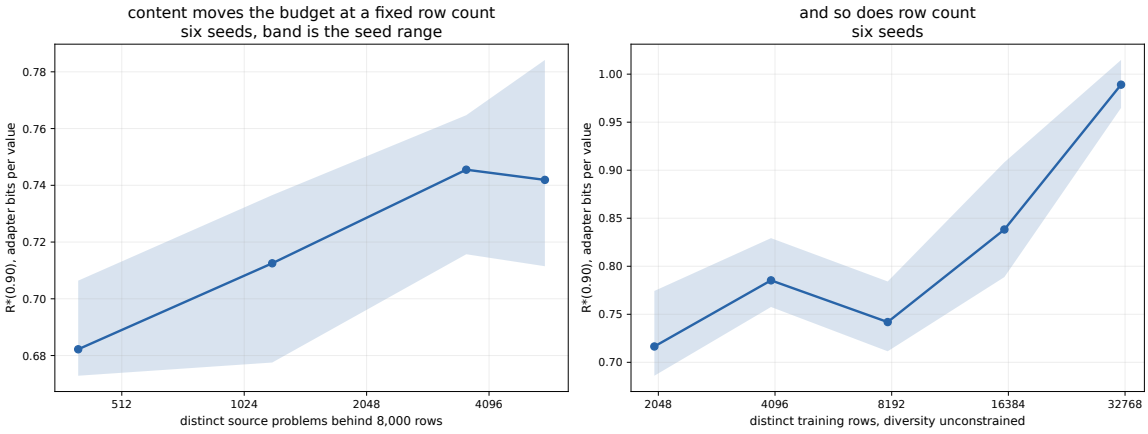


Figure 5: Two related but distinct diversity controls. Left: content diversity changes behind a fixed 8,000-example corpus, with six seeds. Right: the number of distinct training examples changes while optimization exposure is fixed, with six seeds in the updated reduction.

3.6 Known-information control

Since natural corpora do not come with a known number of information bits, a failed correlation could always be blamed on a poor information proxy. To counteract this, we tried building a synthetic mapping task where the source information is known by construction - since we're the ones to construct it. Each prompt maps to one of 16 one-token labels, and the conditions differ in the number of independent four-bit label assignments used to generate the data.

We also estimated the dataset's description length with a conditional *prequential* code: the model is trained on earlier blocks and used to encode later ones through their log loss. A uniform fallback prevents a confidently wrong model from assigning an arbitrarily large code length, and a later switch-code correction lets an entire block fall back to the shared base model when that is cheaper.

After that correction, the prequential code orders the 4-, 64-, 128-, 256-, and 512-bit conditions correctly and stays within 1.1–3.5 times the known source count. The information measure is therefore at least directionally calibrated.

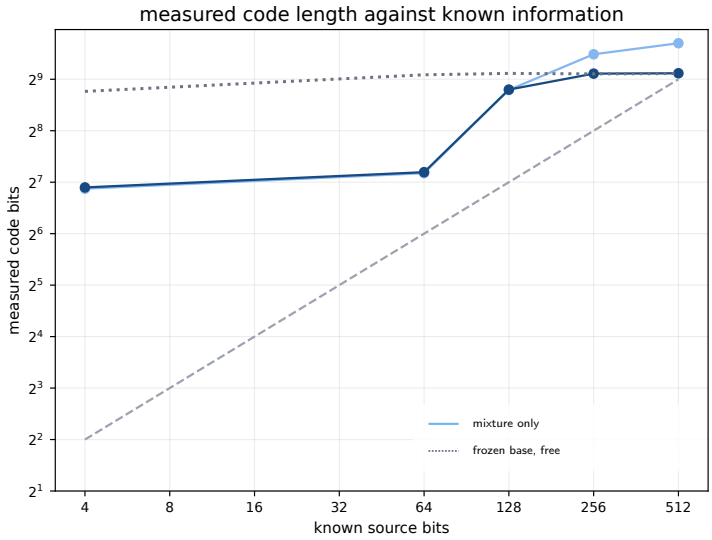


Figure 6: Known-information control. The conditional code recovers the correct order of source information.

3.7 Searching for model-relative dataset information

In alignment with the original internship proposal, the synthetic result suggested that the relevant information might be conditional on what the frozen model already "knows". Before introducing a new model-relative measure, however, we eliminated two simpler explanations: perhaps the rate is just determined by how much the fine-tune improves the model, or perhaps each dataset simply has an intrinsic difficulty.

The first math experiments seemed to support the first idea: across 48 rank-16 runs, held-out prediction gain explained 56% of the variation in $R^\star$. But that relationship covered only a narrow range of gains. We therefore rewrote the same target answers in five deterministic styles, ranging from short plain responses to symbolic ones. These rewrites nearly doubled the learned prediction gain, from 0.555 to 1.062 bits per token, while $R^\star$ moved only from 0.684 to 0.763. A separate 16-fold optimizer-budget sweep changed the amount learned with the content fixed and also failed to produce the expected rate increase. The size of the behavioral improvement is therefore not enough to determine the required rate.

We next changed the frozen model while keeping the fine-tuning setup fixed. The same five corpora were trained on Mistral-7B and Qwen2.5-7B with the same LoRA rank, number

of examples, epochs, and codec. If a corpus had an intrinsic adapter cost, its ordering should remain similar across the two receivers. It did not: only dialogue kept the same position, while instruction tuning, summarization, and math changed substantially in relative cost.

Corpus	Mistral-7B $R^\star(.90)$	Qwen2.5-7B $R^\star(.90)$
Instruction	0.643	0.774
Summarization	0.652	0.388
Math	0.737	0.411
Code	0.891	0.668
Dialogue	0.913	1.176

Table 2: The corpus ordering changes between frozen base models. Each cell averages three independently trained adapters; all 30 rate curves bracket the target.

The earlier math experiments still suggested that content diversity can matter, so we tested that effect on two structured tasks and both base models. We held the total number of training examples fixed while varying the number of text to SQL domains or XBRL tag extraction companies, two finetuning dataset tasks, represented in training. XBRL shows the same direction in all six matched model–seed comparisons; SQL changes much less and inconsistently across seeds.

Task	Model	Groups	Mean $R^\star$, low $\rightarrow$ high	Positive seed pairs
SQL	Mistral-7B	10 $\rightarrow$ 100	0.375 $\rightarrow$ 0.406	2/3
SQL	Qwen2.5-7B	10 $\rightarrow$ 100	0.584 $\rightarrow$ 0.600	2/3
XBRL	Mistral-7B	10 $\rightarrow$ 30	0.260 $\rightarrow$ 0.318	3/3
XBRL	Qwen2.5-7B	10 $\rightarrow$ 30	0.397 $\rightarrow$ 0.469	3/3

Table 3: Fixed-example content-diversity replication. All 36 likelihood-based rate curves bracket $R^\star(.90)$.

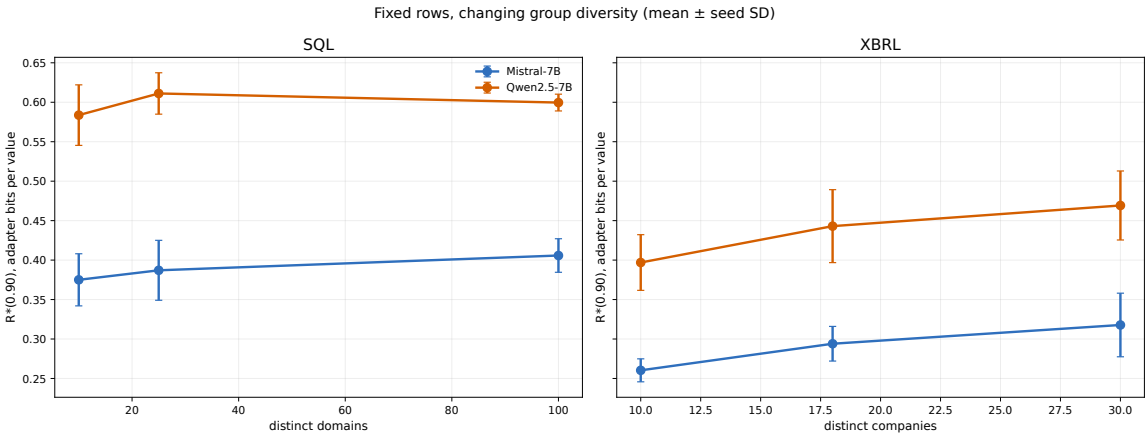


Figure 7: Fixed-example diversity panel. Points show means and bars show one standard deviation over three seeds.

The XBRL intervention is not a clean measure of abstract diversity: increasing the number of companies also changes which companies the adapter sees during training, while the test set covers all companies. The result may therefore partly reflect support coverage. Nearly constant answer entropy rules out a simple label-entropy explanation, but a cleaner diversity intervention is still needed, and one of the things left on our to-do list at the time of writing.

Since by then, the most plausible remaining quantity was clearly the corrections that the dataset asks a particular frozen model to make, we tried to identify a good metric to quantify it. Our most useful family of measures came from the token gradients. For a scored token, the output-head gradient factorizes as

$$g_{it} = h_{it} \otimes (p_{it} - e_{y_{it}}),$$

where h_{it} is the hidden state, p_{it} the model's predicted distribution, and $e_{y_{it}}$ the correct target. Intuitively, one factor describes the representation the model is using and the other describes how its prediction needs to change.

We project these factors to a manageable dimension and summarize the spectrum of their empirical second moment $F = n^{-1}G^\top G$. Our main scalar is

$$I_{\text{corr}}(D; M) = \log\det(I + F) = \sum_j \log(1 + \lambda_j(F)).$$

This quantity grows when the requested corrections are large and spread over many independent directions. Crucially, it depends on both the dataset D and the frozen model M: changing the receiver changes its hidden states, its errors, and therefore the correction spectrum.

After averaging seeds and removing exact data reuses, our comparison contains 22 distinct model–dataset conditions across seven task families. We evaluate each candidate both by how well it orders $R^\star$ within a model and by how well it predicts a task family left out of fitting. Because ten measures were screened on the same data, our reported permutation test also corrected for selecting the best candidate from that set.

Measure	Within-model ρ	Adjusted p	Held-out RMSE	Pooled ρ
Correction Fisher log-volume	**0.764**	**0.00034**	0.187	0.584
Correction Fisher effective rank	0.609	0.0112	**0.181**	**0.814**
Surprise-weighted hidden log-volume	0.614	0.0103	0.204	0.441
Base-model target code length	0.200	0.524	0.261	0.269
Model-only fit	—	—	0.264	—

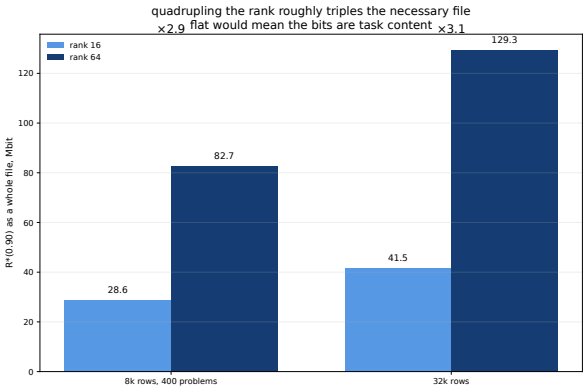
Table 4: Screen of ten model-relative measures against likelihood-based $R^\star(.90)$. RMSE leaves one task family out. ρ is the Spearman correlation, and p is the p-value of the permutation test.

The correction spectrum also follows the earlier SQL/XBRL contrast: its log-volume is nearly flat as SQL diversity increases but rises with XBRL diversity on both base models. It is not simply another name for learning gain; across the distinct conditions, correction log-volume and held-out gain have Spearman correlation -0.275.

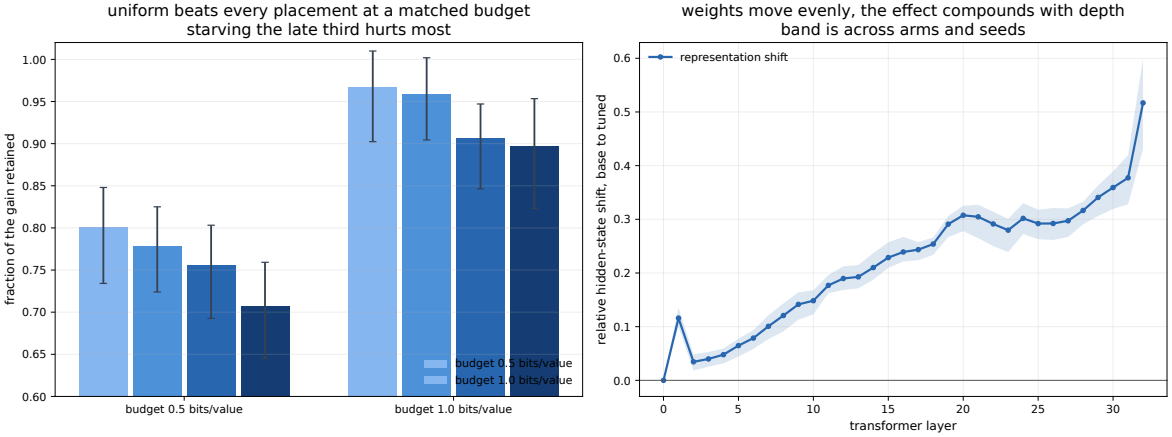
This is our strongest current predictor. It however uses a projected output-head gradient rather than the full LoRA Fisher, and the study contains only two 7B receivers. We therefore treat the correction spectrum as a promising model-relative hypothesis, not yet as an established scaling law. This is ongoing at the time of writing.

3.8 Container and location controls

Finally, we checked whether the adapter architecture itself imposes a large cost independent of task content. Quadrupling LoRA rank from 16 to 64 increases the whole-file $R^\star$ by roughly threefold in two matched conditions, suggesting that a substantial part of the message scales with the carrier. We also asked whether broad layer location could guide bit allocation. At matched 0.5- and 1.0-bit budgets, uniform precision beats starving the early, middle, or late third of the network; starving late layers hurts most. Weight-update RMS is almost flat across depth, while the induced representation shift grows toward the final layers.



(a) Whole-file threshold at ranks 16 and 64.



(b) Layer-band allocation and depth-wise effect.

Figure 8: Container and location controls. Rank scaling is closer to proportional growth than to a fixed task message. At matched total rate, uniform precision beats starving any third of the network; late-layer starvation is most damaging even though update RMS is almost flat across bands.

These controls reinforce the earlier RMSE result: no simple parameter count or global reconstruction norm tells us which bits preserve behavior. The required rate depends both on the adapter used to carry the update and on what the frozen model must change to solve the task.

3.9 Takeaways

The main methodological result of this chapter, which tackles the information theoretic limits of finetuning before its merging with reasoning topics in Section 5 is straightforward: if we want to ask how much information a fine-tune writes, we should measure the size of an actual reloadable update that preserves a stated behavior, rather than infer it from LoRA rank or weight error.

The experiments then rule out several simple predictors of that rate. It is not set by source entropy, base-model loss, or the magnitude of the behavioral improvement alone. The same corpus can require different rates on different frozen models, and content diversity matters in some tasks but not others. The strongest current explanation is model-relative: the rate tracks the number and magnitude of correction directions that the training data asks the frozen model to make. That result is promising but still needs prospective replication.

4 Reasoning trajectories and objective-relative state

4.1 Literature review

4.1.1 Written reasoning and action traces

Chain-of-thought prompting elicits written intermediate steps before an answer and improves several reasoning tasks at sufficient model scale (Wei et al. 2022). ReAct extends the trace with actions and observations, so intermediate text supports both deliberation and interaction with an environment (Yao et al. 2023). These methods establish the practical value of extra computation in the sequence. They do not show that each written step matches a discrete internal update or that the text faithfully records the cause of the answer.

Work on efficient entity tracking makes the state problem more exact. Standard attention needs growing depth for a chain of state changes in the formal setting studied by Fagnou et al. (2024), which includes my tutors, while their modified attention composes the chain more directly.

4.1.2 Hidden trajectories and mechanistic state

The exact processes by which CoT can yield higher problem-solving capabilities is the subject of a lot of ongoing research. Though one can guess that composition of abstract claims is part of it, this has yet to be shown clearly. Mechanistic analyses ask which internal components move or write task information. On fictional ontology tasks, Dutta et al. (2024) report a split between earlier relation transfer and later answer-writing paths. Across abstract rule and analogy tasks, Yang et al. (2025) identify early symbol abstraction, intermediate induction, and late retrieval components. These studies support structured, task-linked computation, but their units come from the chosen task and intervention target.

Trajectory work shifts the unit of analysis from a component to a position or step. Qian et al. (2025) identify "thinking tokens" with high estimated mutual information, Yu et al. (2025) study state-aware changes across reasoning stages, and Sun et al. (2026) report step-specific geometry and correctness signals. The methods differ in target, estimator, and granularity. Their findings motivate a direct test of whether boundaries selected for answer formation, symbolic updates, correctness, and geometric compression agree when the token budget stays fixed.

4.1.3 Latent reasoning and reusable interfaces

Continuous latent reasoning removes the requirement that every intermediate state decode to text. Coconut feeds hidden states back into the model as continuous thoughts and trains their repeated use (Hao et al. 2025). A later preprint uses causal perturbations and adversarial tests to argue that apparent latent reasoning can instead rely on weakly meaningful placeholders in some settings (Zhang et al. 2025). This disagreement makes output quality alone insufficient evidence for a reusable state interface.

Prior work thus finds useful tokens, stages, circuits, and latent states, but each selects an often new notion of state through its task and measure. As we studied the implications of the information theoretic limits of finetuning on RLMs, we came across two clear questions : whether one token decomposition remains useful across several objectives, and whether a decoded or produced state remains sufficient after the source history is removed and the state is used again. We began with the simpler idea that a long verbal derivation might hide a short series of stable updates.

4.2 Trajectory pipeline and first hypotheses

Our initial hypothesis was that a long chain of thought might follow a *solution object*, composed of compact internal updates corresponding to meaningful steps of the reasoning. If such objects existed naturally, several things should happen at roughly the same places in the trace. A meaningful text boundary should coincide with a localized latent change; equivalent computations should permit causal transfer; similar operations should have similar representations; and those units should help predict whether the solution is correct.

We thus tried to identify global, defensible, utilty-neutral partitions in CoTs, which would have allowed us to split CoT streams into chunks, which could then have been classified as different types of reasoning operations.

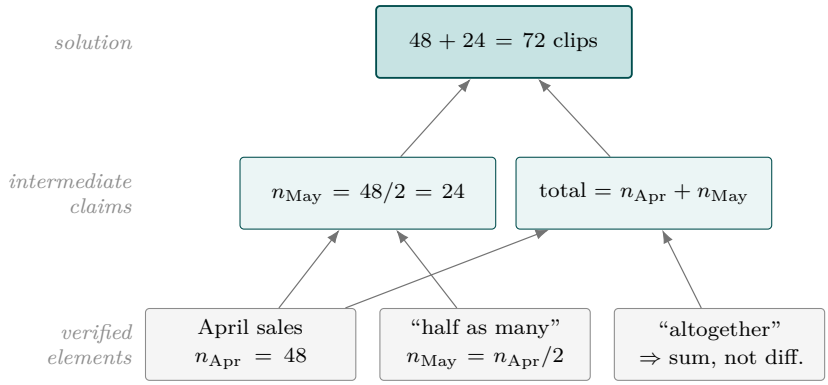


Figure 9: *Problem. Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?*
A hypothetical solution object viewed as a directed acyclic graph over claims. Leaves are directly verifiable readings of the problem statement; internal nodes are operations that consume earlier claims; the sink is the answer.

4.3 No universal partition of the token lattice

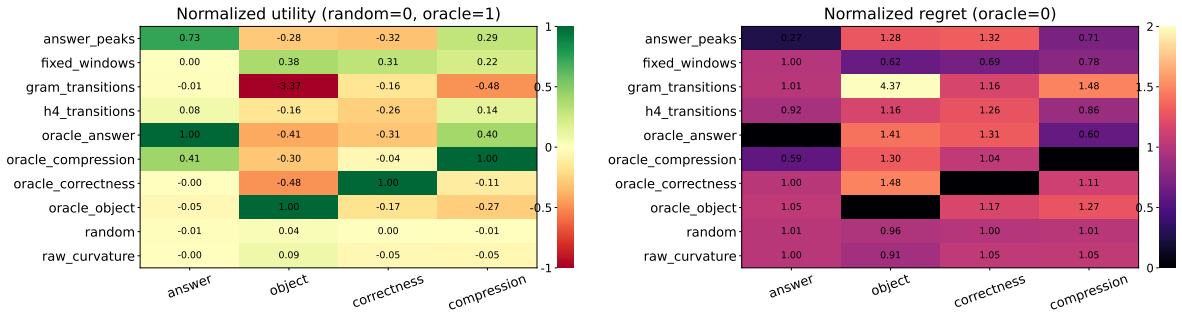
For this comparison, every gap between two generated tokens could be chosen as a boundary. Each trace received the same number of boundaries, based on its number of verified symbolic updates, so one method could not win simply by selecting more points. For each objective, an oracle was allowed to see the true objective scores and choose the best boundaries; random selection was normalized to zero and the oracle to one.

We used four objectives. The first rewards boundaries where the representation moves toward the gold answer; the second rewards completion of a verified symbolic update; the third rewards changes in a held-out correctness probe; and the fourth rewards changes in local PCA geometry, used as a compression-oriented signal. We then asked whether the boundaries that are good for one objective are also good for the others.

They mostly are not. On 310,044 held-out SmolLM3-3B tokens and 61,691 Qwen3-14B tokens, overlap between objective-specific oracle boundaries is low: F1 within four tokens ranges from 0.098 to 0.210 for SmolLM and from 0.130 to 0.294 for Qwen. No single rule performs well on all four objectives: the best worst-objective utility is only 0.020 for SmolLM and 0.080 for Qwen. Sentence boundaries are particularly weak for answer and symbolic-update utility, although raw cosine peaks retain some answer-related signal.

The negative result is not caused by an absence of learnable structure. Supervised detectors recover their own objective's boundaries with held-out ROC–AUC of 0.999/0.984/0.849/0.868 for SmolLM and 0.998/0.992/0.859/0.825 for Qwen (answer, symbolic update, correctness, and compression). In other words, each target leaves a strong enough signal to decode, but

the signals select different partitions of the same trace.



(a) Normalized utility; random is zero and the target oracle is one.

(b) Normalized regret; the target oracle is zero.

Figure 10: Objective-relative partitions on the primary SmolLM corpus.

Our results thus hinted that useful boundaries do exist, but that they are extremely utility-dependent. This shifted the project from searching for one canonical partition to asking what kind of state is sufficient for a specified future use.

4.4 Judged semantic spans

However, our four objectives were deliberately operational and might simply have missed the semantic steps that a human reader would naturally call reasoning. To test this, a larger model used as an LLM-as-a-judge, in this case Qwen-110B-A25B, labeled 415 overlapping windows from concise correct SmolLM traces with nine edit types: *orient*, *bind variable*, *add relation*, *derive value*, *verify*, *correct error*, *plan*, *extract answer*, and *mixed*. After alignment and edge filtering, the audit retained 1,761 labeled spans over 58 traces, with median length 29 tokens. These semantic boundaries are strongly visible in the hidden states. A question-disjoint latent detector reaches AUC 0.982 overall and 0.969 away from sentence endings. However, lexical TF–IDF alone reaches 0.892, showing that much of the type information is also present in the words themselves.

Most importantly, these human-like semantic boundaries still transfer poorly to the four earlier objectives: their utilities are $-0.012/0.043/0.036/0.324$. This result was thus strongly limited, with a readable semantic decomposition being useful, and yet not a unique decomposition privileged by answer formation, symbolic updates, correctness, and compression.

4.5 Exact control: snapshot information is not memory

So far, the experiments concerned how to partition a trace. A separate question is whether a representation that contains the right information *now* remains useful after the system continues operating. A state can be perfectly decodable at one instant and still be a bad memory if its update rule forgets important information later.

To isolate that distinction from language-model noise, we built an exact finite control, in the same fashion as our synthetic finetuning task from Section 3. In this task, a controller observes a short history generated from a finite hidden state and must make decisions over a future horizon using one of several memory representations. Because the hidden-state model, memory update, and planner are all explicit, any difference in performance can be attributed to the representation and its update rule rather than to generation or judging errors.

Across 90 task cells, the full public history and a 12-bit text suffix start with exactly the same support, state mutual information (2.512 bits), and path leakage (1.781 bits). Yet, they never ended up matching in regret: pooled regret with the text suffix is 7.19 times higher. At

horizon 32, regret is 2.20 with full history and 30.91 with the suffix. Both representations contained the same initial observations, but the suffix discarded older information as new observations arrive.

A 12-bit explicit field state performed much better: it removed 79.3% of the regret of having no memory, compared with 93.7% for full history and 54.8% for the text suffix. It still failed when the held-out goal required fields that its training objective did not preserve. This highlighted that snapshot information and update closure are separate requirements.

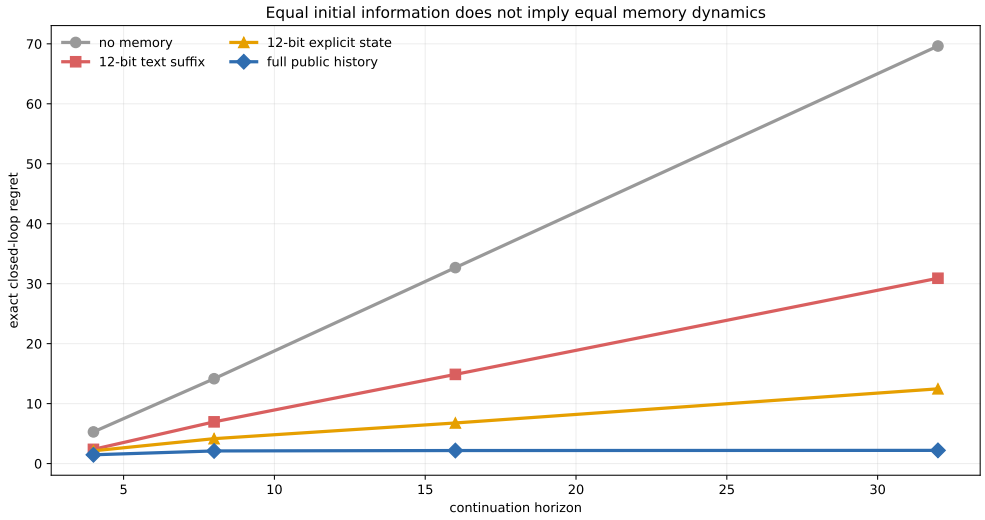


Figure 11: Exact closed-loop regret over continuation horizon.

This control clarified what a reusable state must do: it must summarize the current history well enough for the task, **and** it must remain sufficient after repeated updates.

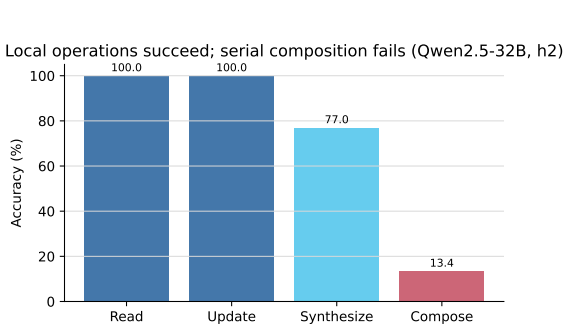
4.6 Native language-model state and explicit handoff

We designed a synthetic addition task where the necessary intermediate state is known exactly. A short program leads to one of eight possible three-bit states, and a separate pointer table maps that state to the final answer. This lets us test four abilities separately: read a state, update it, infer the state reached by a history, and use that inferred state in a final lookup.

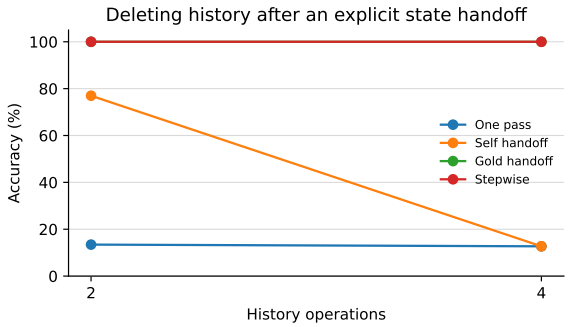
Frozen Qwen2.5-32B can read a supplied state and apply a local update perfectly. At horizon two it also infers the endpoint of the history with 76.98% accuracy. But when asked to infer that endpoint and immediately use it in the final lookup within one prompt, accuracy falls to 13.44%, close to the eight-state chance level.

If we stop after the model predicts the endpoint, delete the original history, and pass only that predicted state to a fresh model call, final accuracy returns to 76.98%. Passing the correct endpoint instead reaches 100%, as does explicit step-by-step decimal execution. The model therefore contains enough information to infer the intermediate state, but the one-pass computation does not reliably route that state into the next operation.

Hidden-state probes also retain information about the path used to reach the endpoint, and the late-layer subspace we tested does not outperform a random subspace control. We therefore do not claim that the frozen model exposes one clean native code that can simply be patched between computations.



(a) Local competence with failed serial composition.



(b) Deleting history after an explicit handoff.

Figure 12: Frozen Qwen2.5-32B state-factorization assay.

4.7 Rate, shared meaning, and closure

Because of this result, instead of continuing to search for an assumed native interface, we supplied an explicit one and asked what properties were necessary for it to work reliably.

To make the state space less trivial, we moved to a finite proof task. The state was in this case a four-bit fact ledger indicating which of four facts are currently true, giving 16 possible states. Horn-style implication rules updated the ledger one deduction at a time. At each step, the previous rules are discarded, and only the current state code is passed to the next model call, so the experiment would only be succesful if the code itself carried the needed state.

We compared four code designs with equal training data and compute. A lossy three-bit code must merge some of the 16 states. A canonical code gives every state one stable name. A five-bit padded code adds unused capacity while keeping one name per state. Finally, a five-bit path-aliased code deliberately gives the same semantic state two possible names depending on how that state was reached. The last pair isolates extra capacity from path-dependent naming.

Canonical and padded codes recover all 16 states in all three seeds and remain exact through the tested horizon of 256. Mean final accuracy is 97.22% for the canonical interface, compared with 72.22% for one-pass reasoning, a paired gain of 25 points with context-bootstrap interval $[12.50, 38.89]$. The three-bit code has one bit of unavoidable residual state uncertainty and never recovers the exact state. Extra unused capacity does not hurt the padded code, but path-dependent naming does: after three active deductions, the aliased code has only 30–40% exact state, versus 100% for the equally large padded code.

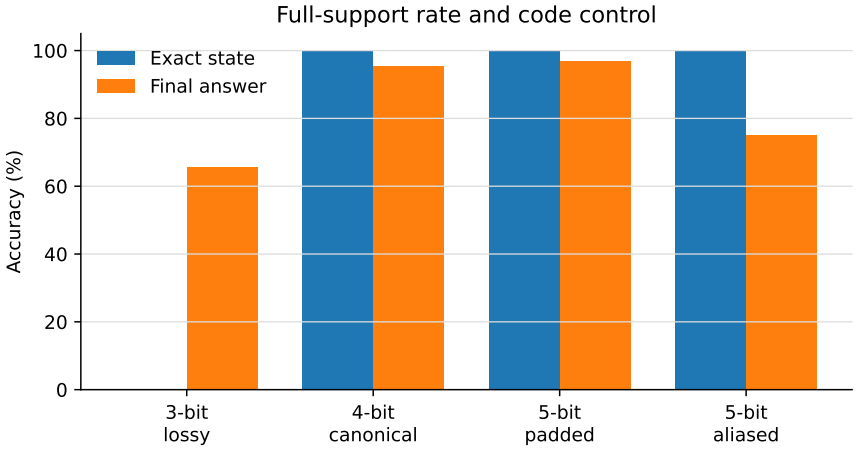


Figure 13: Full-support rate and code control in the later proof-state suite.

As the 5-bit bar of Figure 13 shows, capacity is not automatically better if the extra bits preserve irrelevant history. The padded code has the same size without this ambiguity and stays stable; the aliased code carries path information and loses closure.

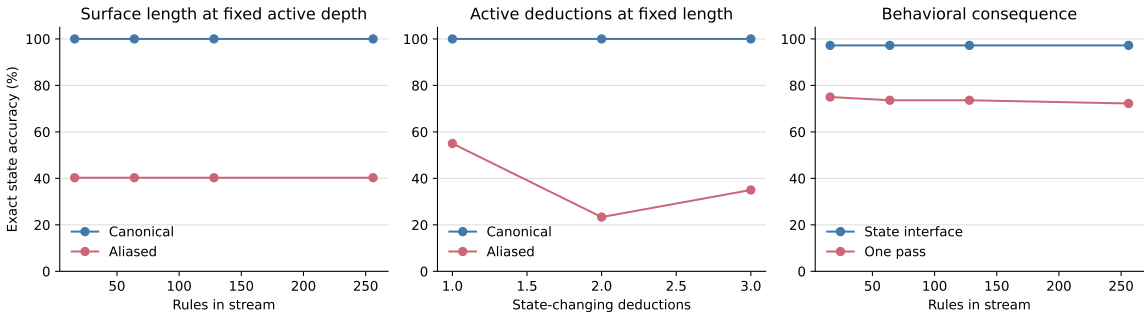


Figure 14: Length, active depth, and behavior in the canonical causal-state suite. Padding the rule stream at fixed active depth leaves exact state unchanged, whereas increasing state-changing deductions damages the aliased code. Canonical final behavior remains high through the padded horizons.

After this, the proof task still had a very regular transition rule, so we finally moved to mixed register programs containing addition, XOR, swap, and conditional updates. Here, the model applies a single update to its current supplied state correctly about 89% of the time. That sounds strong locally, but after roughly 31 self-fed updates the complete horizon-32 accuracy drops to only 7.92%, barely above the 6.25% chance rate; one-pass reasoning reaches 5.83%. This pattern was not model-dependent, and adding a fifth code bit did not repair it.

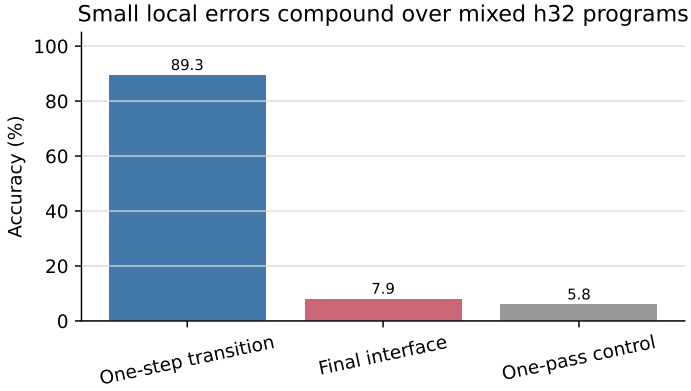


Figure 15: Local versus global reliability on held-out mixed register programs. Conditional transition accuracy is computed from the state actually supplied at each recursive call, so earlier errors do not make a later correct local update appear wrong. Roughly 89% local accuracy still yields 7.9% complete horizon-32 accuracy.

This failure sets the present limit of the interface approach, and more generally of the forced causal state usage. A transition function that is 89% correct in one step is not reliable enough for dozens of recursive uses, a number of steps RLMs can easily reach when making advanced deductions. Small semantic errors compound until the state is nearly useless. Long-horizon reliability therefore has to be measured under self-conditioned use, not inferred from gold-input one-step accuracy.

4.8 Takeaways

The reasoning experiments progressively weakened the original idea of a natural, canonical intermediate state. We found strong latent structure, but different objectives select different

boundaries. We found states that are easy to decode, but that does not mean the model will route them into the next computation. We built explicit compact interfaces that work under controlled conditions, but only when they have enough capacity, one stable meaning per state, and sufficiently accurate self-updates.

More recently, these negative results concerning a discrete, ordinate structure to CoT however oriented us on the path of a more general, continuous view of reasoning, which Paul found a good intuition for that could apply to our initial topic. This is what we're currently working on, and the next chapter will present my ongoing attempt at unifying the two parts of the internship.

5 Information limits in fine-tuning for reasoning

5.1 A measured split between reasoning and answer updates

In our work towards the unification of the two tracks taken during our work so far, we started to study the relative importance and informational weight of the reasoning chain itself versus the final answer. We fine-tuned rank-16 all-linear LoRA adapters on MetaMathQA with two frozen NF4 backbones, Mistral-7B and Qwen2.5-7B. For each model, we trained on the full response, the reasoning span only, or the answer span only.

After training, each fixed adapter was compressed through the same dense codec ladder used in Section 3. We measured the exact serialized file rate in bits per LoRA value and the signed held-out bits per token saved against the frozen model, separately for the reasoning and answer spans. For a frozen model M and span utility U, we write

$$R^{\star}_{M,U}(\tau) = \min_{A} R(A) \quad \text{such that} \quad W_{M,U}(A) \geq \tau W_{M,U}(A_{\text{raw}}), \tag{4}$$

where $W_{M,U}$ is the held-out gain on that span.

On the Mistral model, full-response and reasoning-only supervision save respectively 0.567 and 0.568 bits per reasoning token; on Qwen, they save 0.101 and 0.100. Adding answer supervision therefore seems to have only a negligible effect on the bit rate.

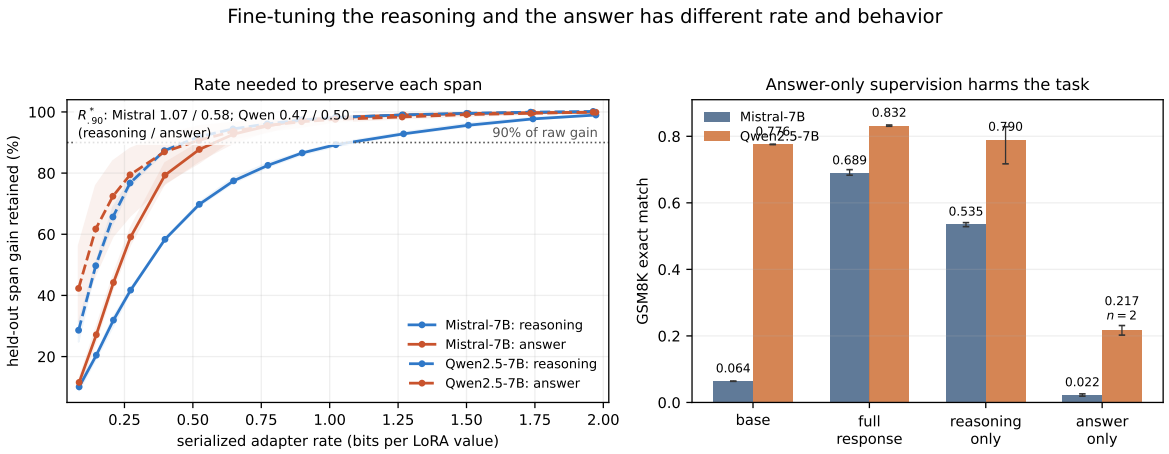


Figure 16: Fine-tuning rate and behavior for the MetaMathQA span-supervision study. Left: held-out gain retained by compressed full-response adapters, averaged over three seeds; shading shows the seed range. Right: mean GSM8K exact match for the raw adapters; error bars show the seed range.

5.2 Initial analysis

This gives a completed reasoning-specific instance of the receiver dependence found in Section 3. A training corpus does not have one adapter rate by itself. The required rate depends on

the frozen model and on which learned behavior must survive compression. It also matches the objective-relative result in Section 4: answer formation, symbolic updates, correctness, and continuation need not select the same representation. At training time, the carrier is a serialized adapter; at inference time, it is a trace partition or state code. In both cases, nominal size alone does not identify the information that supports a stated use.

So far, we thus have evidence for two utility-relative channels. In the training-time channel, the frozen model receives a compressed update and must retain a chosen behavior. In the inference-time channel, a later computation receives a compressed history or state and must retain a chosen use. The experiments show that both channels must be judged by what the receiver can still do.

5.3 Limits of the current bridge

At the time of writing this, I am still in the midst of analyzing the results of the experiment above, which obviously remain preliminary. It uses one training corpus, two base models, one lexical split point, and far fewer held-out answer tokens than reasoning tokens. The experiment measures supervised next-token gain and GSM8K accuracy, so it does not establish how RLVR allocates update rate when only an outcome reward is supplied. Nor does it yet show whether a compressed adapter loses internal reasoning structure before or after it loses answer accuracy.

More broadly, so far, the two halves of the report still meet at the level of a common question rather than one predictive law. Chapter 3 measures how many adapter bits preserve a behavior, while Chapter 4 shows that different future uses select different intermediate structure and that recursive use adds a closure requirement. We have not yet shown that a quantity measured before fine-tuning predicts a reasoning-specific rate, or that changing adapter rate predictably changes the internal state properties measured in the reasoning experiments.

5.4 Promising paths toward a unified result

Intuitively, the highest-upside next step is to make the two tracks predict each other. The correction-spectrum measure from Section 3 can be computed on the frozen model separately for reasoning and answer tokens before training. If those utility-specific spectra predict the later $R^{\star}_{M,U}$ values across new models and datasets, the current model-relative account would become a predictive statement about which parts of a reasoning task are expensive to write.

Another path forward would be to compress a fixed reasoning adapter, and follow several properties along the same rate ladder. We could measure its objective-specific boundaries, operation decoding, trajectory geometry, and explicit state handoff tests from Section 4. If these properties fail at reproducibly different rates, we would have direct evidence that even within one trained model there is no single behavior-independent "amount of reasoning" stored in the update. If they fail together, that would instead identify a much stronger common bottleneck.

There remains a lot of paths forward, and as for research in general, my success will likely depend on my capacity to ask the right question and discern between the most promising hypothesises.

6 Conclusion

The internship started from a broad question: how much information does a model have to acquire, or to keep, for a learned behavior to survive? The first obstacle was that "information" is not something one can simply go and measure. Much of our work since April consisted in

trading intuitive proxies for computable proxies, e.g., a serialized adapter file, a held-out gain on a named span, a boundary set scored against a stated objective.

A fair share of what came out of those measurements was negative. Some of it was expected, in the form of controls we ran for rigor and whose failure would have been surprising. Weight reconstruction error does not predict behavioral preservation. The code length of the source does not determine $R^\star$. Neither does the size of the improvement the fine-tune buys, since XBRL produced the largest gain of the campaign at the lowest measured rate, and the ordering of the five corpora rearranged itself as soon as the frozen base model changed. Of course, not all of our negative results were expected, unfortunately. On the reasoning side, no single partition of the token lattice served answer formation, symbolic updates, correctness and compression at once. A state could be decoded cleanly and still not be routed into the next operation. As it turned out, a transition rule that is right 89% of the time in one step turns out to be rather useless if used a high number of times in a single reasoning chain.

I do think it was useful to practice going from an observed phenomenon, to a hypothesis, to an experiment that verifies said hypothesis. Nearly every course correction visible in this report came out of a result that did not work.

The two tracks converged on the same idea: useful information is not an intrinsic property of a dataset or a reasoning trace. It is relative to a receiver, a task, and, when reused recursively, an update rule. On the fine-tuning side, model-relative correction spectra are our strongest predictor of adapter rate; on the reasoning side, useful intermediate structure exists, but the structure we recover depends strongly on what it must be used for.

As stated previously, this internship is still ongoing. By the end of September, I hope to join these two lines more tightly, extend the strongest results to new models and tasks, and turn them into a stronger thesis and potential ICLR submissions. I am very grateful to Paul Caillon and Alexandre Allauzen for their supervision, discussions, and freedom to pursue the questions that emerged throughout the internship.

AI Usage Disclosure

Throughout the internship, OpenAI's harness Codex assisted me in the replication of existing papers, in first drafts of code implementations of well defined research ideas and in debugging. AI output was never used to replace my judgment.

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